

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

This Safety Data Sheet adheres to the standards and regulatory requirements of Mexico and may not meet the regulatory requirements in other countries.

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product identifier

Product name ALSTAR DUAL® 300 SC

Other means of identification

Product code 50002337

Product Registration Number RSCO-MEZC-1101N-0232-064-27.45

Recommended use of the chemical and restrictions on use

Recommended use Insecticide

Restrictions on use Use as recommended by the label.

Manufacturer or supplier's details

Manufacturer FMC AGROQUÍMICA DE MÉXICO,
S. DE R.L. DE C.V. AV. VALLARTA NO.
6503, LOCAL A1-6, COL. CD. GRANJA,
45010 ZAPOPAN, JALISCO, MÉXICO
TEL.: (33) 3003 4500
CONTACTOMEXICO@FMC.COM
SDS-Info@fmc.com

Emergency telephone

For leak, fire, spill or accident emergencies, call:
800-681-9531 (CHEMTREC - Mexico)
1 703 / 741-5970 (CHEMTREC - International)

Medical emergency:
911
SINTOX (Toxicological Information Service): 800 009 2800; 55
5611 2634 and 55 5598 6659, service 24 hours a day, 365
days a year.

SECTION 2. HAZARDS IDENTIFICATION

Classification of the hazardous chemical or mixture according to the GHS and NOM-018-STPS-2015

Flammable liquids : Category 4

Acute toxicity (Oral) : Category 4

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Acute toxicity (Inhalation) : Category 4

Skin sensitization : Category 1

Specific target organ toxicity : Category 1 (Central nervous system)
- single exposure

Specific target organ toxicity : Category 1 (Central nervous system)
- repeated exposure

Label elements, including precautionary statements and hazard pictograms according to the GHS and NOM-018-STPS-2015

Hazard pictograms :



Signal Word : DANGER

Hazard Statements : H227 Combustible liquid.
H302 + H332 Harmful if swallowed or if inhaled.
H317 May cause an allergic skin reaction.
H370 Causes damage to organs (Central nervous system).
H372 Causes damage to organs (Central nervous system)
through prolonged or repeated exposure.

Precautionary Statements :

Prevention:

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

P260 Do not breathe mist or vapors.

P264 Wash skin thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P271 Use only outdoors or in a well-ventilated area.

P272 Contaminated work clothing should not be allowed out of the workplace.

P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell. Rinse mouth.

P302 + P352 IF ON SKIN: Wash with plenty of water.

P304 + P340 + P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell.

P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.

P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.

P362 + P364 Take off contaminated clothing and wash it before reuse.

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 Store in a well-ventilated place.

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards

Very toxic to aquatic life with long lasting effects.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
imidacloprid (ISO)	138261-41-3	>= 20 -< 30
Bifenthrin	82657-04-3	>= 1 -< 5

SECTION 4. FIRST AID MEASURES

- | | |
|---|---|
| General advice | : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended. |
| If inhaled | : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician. |
| In case of skin contact | : Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use. |
| In case of eye contact | : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist. |
| If swallowed | : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital. |
| Most important symptoms and effects, both acute and delayed | : Harmful if swallowed or if inhaled.
May cause an allergic skin reaction.
Causes damage to organs. |

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



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Causes damage to organs through prolonged or repeated exposure.
Contact with skin may cause tingling, itching, burning, or numbness at the site of contact. Inhalation may irritate the nose, throat, and lungs. Swallowing large quantities may result in throat irritation, nausea, abdominal pain, and vomiting. Signs of exposure are uncoordinated gait, tremors, and reduced activity. Very high oral exposures may lead to lethargy, vomiting, diarrhea, salivation, muscle weakness and ataxia.

Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.

Notes to physician : In case of poisoning, call the emergency numbers SINTOX (control center of intoxications): 800 009 2800; (55) 5611 2634 and (55) 5598 6659, 24-hour service on 365 days of the year. For emergencies: 911. Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.

Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Halogenated compounds
Carbon oxides
Nitrogen oxides (NO_x)
Fluorinated compounds
Chlorinated compounds
Hydrogen chloride
Hydrogen fluoride

Specific extinguishing methods : Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment for fire-fighters : Firefighters should wear protective clothing and self-contained breathing apparatus.

SAFETY DATA SHEET

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ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

SECTION 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Evacuate personnel to safe areas.
Ensure adequate ventilation.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
- Environmental precautions : Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.
- Methods and materials for containment and cleaning up : Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal.
- For further cleaning instructions call CHEMTREC, 800-681-9531.

SECTION 7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition.
- Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- For incompatible materials see section 10.
- Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.
When using do not eat or drink.

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

When using do not smoke.
Wash hands before breaks and at the end of workday.

Conditions for safe storage : No smoking.
Keep in a well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.

Hand protection
Material : Protective gloves

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Plan first aid action before beginning work with this product.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Form : suspension

Color : No data available

Odor : No data available

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version 3.0	Revision Date: 26.03.2026	SDS Number: 50002337	Date of last issue: - Date of first issue: 13.11.2019
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Odor Threshold	:	No data available
pH	:	5.67 (25 °C) Concentration: 10 g/l
Melting point/ range	:	No data available
Boiling point/boiling range	:	No data available
Flash point	:	85 - 88 °C
Evaporation rate	:	No data available
Self-ignition	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	No data available
Density	:	1.1142 g/cm ³ (20 °C)
Solubility(ies) Water solubility	:	Miscible
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	No data available
Molecular weight	:	Not applicable
Other relevant data		
Explosive properties	:	Not explosive

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Oxidizing properties : Non-oxidizing

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: No decomposition if stored and applied as directed.
Possibility of hazardous reactions	: Vapors may form explosive mixture with air. No decomposition if stored and applied as directed.
Conditions to avoid	: Avoid extreme temperatures. Avoid formation of aerosol. Heat, flames and sparks.
Incompatible materials	: Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	: No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute toxicity

Harmful if swallowed or if inhaled.

Product:

Acute oral toxicity	: Acute toxicity estimate: 1,010.7 mg/kg Method: Calculation method Remarks: NOM-232-SSA1-2009 Acute toxicity estimate: 1,457 mg/kg Method: Calculation method
Acute inhalation toxicity	: Acute toxicity estimate: 3.643 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: Calculation method Assessment: The component/mixture is moderately toxic after short term inhalation. Remarks: NOM-232-SSA1-2009
Acute dermal toxicity	: Acute toxicity estimate: 7,061.25 mg/kg Method: Calculation method Remarks: NOM-232-SSA1-2009 Acute toxicity estimate: > 5,000 mg/kg Method: Calculation method

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Components:

imidacloprid (ISO):

- Acute oral toxicity : LD50 (Rat, male and female): > 1,000 mg/kg
Symptoms: Tremors, piloerection, Breathing difficulties
Remarks: no mortality
- LD50 (Rat, female): 300 - 2,000 mg/kg
Method: OECD Test Guideline 423
Symptoms: Fatality, Convulsions, piloerection
GLP: yes
Assessment: The component/mixture is moderately toxic after single ingestion.
- LD50 (Rat, female): 300 - 2,000 mg/kg
Method: OECD Test Guideline 420
Symptoms: Fatality, Tremors, ataxia
GLP: yes
Assessment: The component/mixture is moderately toxic after single ingestion.
- LD50 (Rat, female): ca. 2,567 mg/kg
Method: OECD Test Guideline 425
Symptoms: Fatality, Breathing difficulties
GLP: yes
- Acute inhalation toxicity : LC50 (Rat, male and female): > 5.31 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: no mortality
- LC50 (Rat, male and female): 5.17 mg/l
Exposure time: 4 h
Method: OECD Test Guideline 403
Symptoms: hypoactivity
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: no mortality
- LC50 (Rat, male and female): > 4.9 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Breathing difficulties, ataxia, Convulsions, Tremors
Assessment: The component/mixture is minimally toxic after short term inhalation.
- Acute dermal toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 402
Symptoms: Irritation

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: no mortality

LD50 (Rabbit): > 2,000 mg/kg

Bifenthrin:

Acute oral toxicity : LD50 (Rat, male and female): 50.2 - 58.8 mg/kg
Symptoms: Convulsions, Tremors

Acute inhalation toxicity : LC50 (Rat, female): 0.6 - 1.2 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors, Convulsions

LC50 (Rat, male): 1.10 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors, Fatality

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Remarks: no mortality

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Assessment : Not classified as irritant
Result : slight or no skin irritation.

Remarks : May cause skin irritation and/or dermatitis.

Remarks : May cause skin irritation and/or dermatitis.

Components:

imidacloprid (ISO):

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
GLP : yes

Bifenthrin:

Species : Rabbit
Method : OECD Test Guideline 404
Result : slight or no skin irritation.
GLP : yes

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	: Rabbit
Result	: Slight or no eye irritation
Assessment	: Not classified as irritant

Remarks	: Vapors may cause irritation to the eyes, respiratory system and the skin.
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Remarks	: Vapors may cause irritation to the eyes, respiratory system and the skin.
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Components:

imidacloprid (ISO):

Species	: Rabbit
Result	: No eye irritation
Method	: OECD Test Guideline 405
GLP	: yes

Bifenthrin:

Species	: Rabbit
Result	: Slight or no eye irritation
Method	: OECD Test Guideline 405
GLP	: yes

Respiratory or skin sensitization

Skin sensitization

May cause an allergic skin reaction.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Assessment	: May cause sensitization by skin contact.
Result	: May cause sensitization by skin contact.

Remarks	: Causes sensitization.
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Remarks	: Causes sensitization.
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Remarks	: Causes sensitization.
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Components:

imidacloprid (ISO):

Test Type	: Maximization Test
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SAFETY DATA SHEET

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ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Species : Guinea pig
Result : Does not cause skin sensitization.

Test Type : Local lymph node assay (LLNA)
Species : Mouse
Method : OECD Test Guideline 429
Result : Does not cause skin sensitization.
GLP : yes

Bifenthrin:

Test Type : Maximization Test
Routes of exposure : Skin contact
Species : Guinea pig
Method : OECD Test Guideline 406
Result : May cause sensitization by skin contact.
GLP : yes

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Result: negative
GLP: yes

Genotoxicity in vivo : Test Type: Cytogenetic assay
Species: Chinese hamster
Result: negative
GLP: yes

Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

GLP: yes

Test Type: dominant lethal test
Species: Mouse
Result: negative

Test Type: chromosome aberration assay
Species: Mouse
Result: negative

Bifenthrin:

Genotoxicity in vitro

: Test Type: Ames test
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: gene mutation test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes

Test Type: Mouse lymphoma assay
Metabolic activation: with and without metabolic activation
Result: negative

Genotoxicity in vivo

: Test Type: Sex-linked Recessive Lethal Test
Species: Drosophila melanogaster (vinegar fly)
Result: negative

Test Type: unscheduled DNA synthesis assay
Species: Rat
Method: OECD Test Guideline 486
Result: negative

Test Type: Mammalian erythrocyte micronucleus test (in vivo cytogenetic assay)
Species: Mouse (male and female)
Method: OECD Test Guideline 474
Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

Bifenthrin:

Species	: Rat, female
Application Route	: Oral
Exposure time	: 2 Years
NOAEL	: 3 mg/kg bw/day

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Result	: negative
Species	: Mouse, male
Application Route	: Oral
Exposure time	: 18 month(s)
NOAEL	: 7.6 mg/kg bw/day
Result	: positive
Symptoms	: malignant tumors

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

Effects on fertility	: Method: OECD Test Guideline 416 Result: Animal testing did not show any effects on fertility.
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Method: OECD Test Guideline 416
Result: No effects on fertility and early embryonic development were detected.

Effects on fetal development	: Species: Rabbit Application Route: Oral Dose: 0, 8, 24, 72 mg/kg bw/day General Toxicity Maternal: NOAEL: 8 mg/kg bw/day Method: OECD Test Guideline 414 Result: No teratogenic effects. GLP: yes Species: Rat Dose: 0, 10, 30, 100 mg/kg bw/day General Toxicity Maternal: NOEL: 10 mg/kg bw/day Embryo-fetal toxicity.: NOEL: 30 mg/kg bw/day Method: OECD Test Guideline 414 GLP: yes Test Type: Multi-generation study Species: Rat Application Route: Oral Dose: 8, 20, 56 mg/kg bw/day General Toxicity Maternal: NOEL: 20 mg/kg body weight Developmental Toxicity: NOEL: 20 mg/kg body weight Result: No teratogenic effects. GLP: yes
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Bifenthrin:

Effects on fertility	: Test Type: Two-generation study Species: Rat Application Route: Oral General Toxicity Parent: NOAEL: 3 mg/kg bw/day General Toxicity F1: NOAEL: 5 mg/kg bw/day Result: negative
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SAFETY DATA SHEET

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ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Effects on fetal development : Test Type: Embryo-fetal development
Species: Rabbit
Application Route: Oral
General Toxicity Maternal: NOAEL: 2.7 mg/kg bw/day
Teratogenicity: NOAEL: 2.7 mg/kg bw/day
Symptoms: Maternal effects.
Result: No teratogenic effects.

Test Type: Embryo-fetal development
Species: Rat
Application Route: Oral
General Toxicity Maternal: NOAEL: 1 mg/kg bw/day
Teratogenicity: NOAEL: 2 mg/kg bw/day
Result: No teratogenic effects.

Species: Rat
Application Route: Oral
General Toxicity Maternal: LOAEL: 7.2 mg/kg bw/day
Developmental Toxicity: LOAEL: 7.2 mg/kg bw/day
Embryo-fetal toxicity.: NOEL: 9.0 mg/kg bw/day
Method: OECD Test Guideline 426
Result: Animal testing did not show any effects on fertility.,
Some evidence of adverse effects on development, based on
animal experiments.

STOT-single exposure

Causes damage to organs (Central nervous system).

Components:

Bifenthrin:

Target Organs	: Central nervous system
Assessment	: Causes damage to organs.

STOT-repeated exposure

Causes damage to organs (Central nervous system) through prolonged or repeated exposure.

Components:

Bifenthrin:

Target Organs	: Central nervous system
Assessment	: The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Repeated dose toxicity

Components:

imidacloprid (ISO):

Species	: Dog
NOEL	: 1200 ppm
Application Route	: Oral - feed

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Exposure time : 90 d
Method : OECD Test Guideline 409
GLP : yes

Species : Dog
LOAEL : 49 mg/kg
Application Route : Oral - feed
Exposure time : 28 d
Dose : 0, 7.3, 31, 49 mg/kg bw/day
Method : OECD Test Guideline 409
Symptoms : Tremors, ataxia, Vomiting

Species : Dog, male and female
NOEL : 72 mg/kg bw/day
Application Route : Oral - feed
Exposure time : 52 w
Dose : 0, 6.1, 15, 41, 72 mg/kg bw/day
GLP : yes

Bifenthrin:

Species : Rat, male and female
NOEL : 100 ppm
Application Route : Oral - feed
Exposure time : 90 d
Remarks : No toxicologically significant effects were found.

Species : Dog, male and female
NOEL : 2.5 mg/kg bw/day
Application Route : Oral - feed
Exposure time : 13 w
Symptoms : Tremors

Species : Dog, male and female
NOEL : 1.5 mg/kg
Application Route : Oral
Exposure time : 52 w
Dose : 0.75, 1.5, 3, 5 mg/kg bw/day
GLP : yes
Symptoms : Tremors

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

The substance does not have properties associated with aspiration hazard potential.

Bifenthrin:

The substance does not have properties associated with aspiration hazard potential.

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Further information

Product:

Remarks : No data available

Remarks : No data available

Remarks : No data available

Components:

imidacloprid (ISO):

Remarks : No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

imidacloprid (ISO):

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): > 105 mg/l
Exposure time: 96 h
Test Type: static test
Method: EPA OPP 72-1
GLP: yes

LC50 (Salmo gairdneri): 158 - 281 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Oncorhynchus mykiss (rainbow trout)): > 83 mg/l
Exposure time: 96 h
Test Type: static test
Method: EPA OPP 72-1
GLP: yes

LC50 (Cyprinodon variegatus (sheepshead minnow)): 161 mg/l
Exposure time: 96 h
Test Type: static test
GLP: yes

LC50 (Leuciscus idus (Golden orfe)): 178 - 316 mg/l
Exposure time: 96 h
Test Type: static test
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 85 mg/l
Exposure time: 48 h

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Method: US EPA Test Guideline OPP 72-2
GLP: yes

EC50 (*Americamysis bahia* (mysid shrimp)): 0.0341 mg/l
Exposure time: 96 h
Test Type: flow-through test
Method: US EPA Test Guideline OPP 72-3
GLP: yes

LC50 (*Hyalella azteca* (Amphipod)): 0.526 mg/l
Exposure time: 96 h
Method: US EPA Test Guideline OPP 72-2
GLP: yes

NOEC (*Crassostrea virginica* (atlantic oyster)): 23.3 mg/l
Exposure time: 96 h
Method: US EPA Test Guideline OPP 72-3
GLP: yes

EC50 (*Hyalella azteca* (Amphipod)): 0.055 mg/l
End point: Immobilization
Exposure time: 48 h
Method: EPA OPP 72-2

Toxicity to algae/aquatic plants

: EbC50 (*Scenedesmus subspicatus*): > 10 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

NOEC (*Scenedesmus capricornutum* (fresh water algae)): > 119 mg/l
Exposure time: 5 d
Method: US EPA Test Guideline OPP 122-2 & 123-2

Toxicity to fish (Chronic toxicity)

: NOEC (*Salmo gairdneri*): 28.5 mg/l
Exposure time: 21 d
Method: OECD Test Guideline 204
GLP: yes

NOEC (*Oncorhynchus mykiss* (rainbow trout)): 9.8 mg/l
End point: Growth
Exposure time: 98 d
Test Type: Early Life-Stage
Method: US EPA Test Guideline OPP 72-4
GLP: yes

NOEC (*Oncorhynchus mykiss* (rainbow trout)): 9.02 mg/l
End point: Hatching success
Test Type: flow-through test
Method: OECD Test Guideline 210
GLP: yes

Toxicity to daphnia and other aquatic invertebrates (Chronic)

: NOEC (*Daphnia magna* (Water flea)): 1.8 mg/l
Exposure time: 21 d

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

ic toxicity)

Test Type: semi-static test
Method: US EPA Test Guideline OPP 72-4
GLP: yes

EC10 (*Chironomus riparius* (harlequin fly)): 0.00209 mg/l
Exposure time: 28 d

NOEC (*Chironomus tentans*): 0.67 µg/l
End point: Growth
Exposure time: 10 d
Test Type: Static renewal test
GLP: yes

NOEC (*Gammarus pulex*): 0.064 mg/l
End point: Swimming behavior
Exposure time: 28 d
Test Type: static test
Method: OECD 219
GLP: yes

Toxicity to microorganisms : IC50 (activated sludge): > 10000

Toxicity to soil dwelling organisms : LC50 (*Eisenia fetida* (earthworms)): 10.7 mg/kg dry weight (d.w.)
Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (*Coturnix japonica* (Japanese quail)): 31 mg/kg

LD50 (*Coturnix japonica* (Japanese quail)): 2,225 ppm
Exposure time: 5 d

LD50 (*Apis mellifera* (bees)): 0.0037 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity

LD50 (*Apis mellifera* (bees)): 0.0081 µg/bee
Exposure time: 48 h

Ecotoxicology Assessment

Other organisms relevant to the environment : Harmful to bees.

Bifenthrin:

Toxicity to fish : LC50 (*Salmo gairdneri*): 0.00015 mg/l
Exposure time: 96 h
Test Type: flow-through test

LC50 (*Lepomis macrochirus* (Bluegill sunfish)): 0.00035 mg/l
Exposure time: 96 h
Test Type: flow-through test

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

		LC50 (Oncorhynchus mykiss (rainbow trout)): 0.000256 mg/l Exposure time: 96 h Test Type: semi-static test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Pimephales promelas (fathead minnow)): 0.000234 mg/l Exposure time: 96 h Test Type: semi-static test Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia): 0.00011 mg/l Exposure time: 48 h LC50 (Daphnia): 0.0016 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	:	EC50 (algae): 0.822 mg/l Exposure time: 72 h
Toxicity to fish (Chronic toxicity)	:	NOEC (Oncorhynchus mykiss (rainbow trout)): 0.00012 mg/l Exposure time: 21 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 0.0013 µg/l Exposure time: 21 d NOEC (Daphnia magna (Water flea)): 0.00095 µg/l Exposure time: 21 d
Toxicity to soil dwelling organisms	:	LD50 (Eisenia fetida (earthworms)): > 16 mg/kg Exposure time: 14 d Method: OECD Test Guideline 216 Remarks: No significant adverse effect on Nitrogen mineralization.
Toxicity to terrestrial organisms	:	LD50 (Colinus virginianus (Bobwhite quail)): 1,800 mg/kg LD50 (Anas platyrhynchos (Mallard duck)): > 2,150 mg/kg LD50 (Apis mellifera (bees)): 0.07 µg a.i./bee Exposure time: 48 h End point: Acute oral toxicity Method: OECD Test Guideline 213 LD50 (Apis mellifera (bees)): 0.0201 µg a.i./bee Exposure time: 48 h End point: Acute contact toxicity Method: OECD Test Guideline 214

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

ED50 (Apis mellifera (bees)): 143 ng ai larva
Exposure time: 7 d
End point: Acute larval toxicity
Method: OECD 237

ED50 (Apis mellifera (bees)): 15.15 ng ai larva/day
Exposure time: 22 d
End point: Chronic larval toxicity
Method: OECD 239

NOAEL (Apis mellifera (bees)): 2.30 ng ai larva/day
Exposure time: 22 d
End point: Chronic larval toxicity
Method: OECD 239

ED50 (Apis mellifera (bees)): 300 ng a.i/bee/day
Exposure time: 10 d
End point: Chronic oral toxicity
Method: OECD TG 245

NOAEL (Apis mellifera (bees)): 170 ng a.i/bee/day
Exposure time: 10 d
End point: Chronic oral toxicity
Method: OECD TG 245

Persistence and degradability

Components:

imidacloprid (ISO):

Biodegradability : Result: Not readily biodegradable.

Bifenthrin:

Biodegradability : Result: Not readily biodegradable.

Stability in water : Degradation half life (DT50): 2.2 d
Hydrolysis: at 60 °C

Degradation half life (DT50): 15.6 d
Hydrolysis: at 40 °C

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

imidacloprid (ISO):

Bioaccumulation : Remarks: Low potential for bioaccumulation

Partition coefficient: n- : log Pow: 0.33 (20 °C)

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

octanol/water

Method: OECD Test Guideline 107

Bifenthrin:

Bioaccumulation : Species: *Lepomis macrochirus* (Bluegill sunfish)
Bioconcentration factor (BCF): 1,709
Remarks: Due to the distribution coefficient n-octanol/water, accumulation in organisms is possible.

Partition coefficient: n-octanol/water : log Pow: 6.6

Mobility in soil

Components:

imidacloprid (ISO):

Distribution among environmental compartments : Koc: 109 - 411
Remarks: Mobile in soils

Bifenthrin:

Distribution among environmental compartments : Koc: 236610 ml/g, log Koc: 5.37
Remarks: immobile

Stability in soil :

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

Endocrine disrupting properties

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : Appropriate personal protective equipment, as described in Sections 7 and 8, should be worn when handling materials for waste disposal.

The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Containers must be disposed of in accordance with local, state and federal regulations. It is prohibited to reuse, bury,

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

burn or sell containers. Washable containers: Triple wash containers smaller than 20 liters and pressure wash containers of 20 liters or more. Triple wash: Add water up to $\frac{1}{4}$ of the container's capacity, close and shake for 30 seconds. Pour the wash water into the mixing tank, considering this volume of water within the recommended volume for mixing. Perform this procedure three times. Pressure washing: Activate the pressure washing device for 30 seconds, considering the volume of water used as part of the recommended volume for the mixture. For both procedures, make the container unusable by piercing it at the base without damaging the label. Non-washable containers: Containers that cannot be washed, make them unusable by perforating them without damaging the label. In all cases, deliver the containers to collection points indicated by the local container collection program. For more information on the Empty Pesticide Container Management Plan, visit <http://campolimpio.org.mx/>.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number	: UN 3082
Proper shipping name	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Bifenthrin, Imidacloprid)
Class	: 9
Packing group	: III
Labels	: 9
Environmentally hazardous	: yes

IATA-DGR

UN/ID No.	: UN 3082
Proper shipping name	: Environmentally hazardous substance, liquid, n.o.s. Environmentally hazardous substance, liquid, n.o.s. (Bifenthrin, Imidacloprid)
Class	: 9
Packing group	: III
Labels	: Miscellaneous
Packing instruction (cargo aircraft)	: 964
Packing instruction (passenger aircraft)	: 964
Environmentally hazardous	: yes

IMDG-Code

UN number	: UN 3082
Proper shipping name	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Bifenthrin, Imidacloprid)
Class	: 9
Packing group	: III
Labels	: 9
EmS Code	: F-A, S-F
Marine pollutant	: yes

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Domestic regulation

NOM-002-SCT

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Bifenthrin, Imidacloprid)
Class	:	9
Packing group	:	III
Labels	:	9
Environmentally hazardous	:	yes

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

This document has been prepared in accordance with the Globally Harmonized System (GHS). The document consists of 16 points that cover the Official Mexican STANDARD NOM-018-STPS-2015 Harmonized system for the identification and communication of hazards and risks due to dangerous chemical substances in the workplace. 271000

NOM-165-SEMARNAT-2013, Norm establishing a list of substances subject to report for the Registry of Emissions and Pollutant Transfer

Components	CAS-No.	MPU (kg/year)	Transfer/Release (kg/year)
imidacloprid (ISO)	138261-41-3	5000 kg/year	500 kg/year
Bifenthrin	82657-04-3	2500 kg/year	100 kg/year

MPU: Applicable reporting threshold when the substance, pure or in mixture in a composition of more than 1% by weight, is used for industrial activities at facilities that are subject to report or are produced by them

Federal Law for the control of chemical precursors, : Not applicable
essential chemical products and machinery for producing capsules, tablets and pills.

The ingredients of this product are reported in the following inventories:

TW TCSI	:	Not in compliance with the inventory
US TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AU AIIC	:	Not in compliance with the inventory
CA. DSL	:	This product contains the following components that are not on the Canadian DSL nor NDSL.

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

imidacloprid (ISO)

Bifenthrin

Polycarboxylic acid

Organic Acid

JP ENCS	:	Not in compliance with the inventory
JP ISHL	:	Not in compliance with the inventory
KR KECI	:	Not in compliance with the inventory
PH PICCS	:	Not in compliance with the inventory
CN IECSC	:	Not in compliance with the inventory
NZ NZIoC	:	Not in compliance with the inventory
TH TECI	:	Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

Revision Date	:	26.03.2026
Date format	:	dd.mm.yyyy

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardization; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organization for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MERCOSUR - The Agreement for the Facilitation of the Transport of Dangerous Goods; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand

SAFETY DATA SHEET

according to the Mexican Official Standard NOM-018-STPS-2015



ALSTAR DUAL® 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
3.0	26.03.2026	50002337	Date of first issue: 13.11.2019

Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorization and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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End of Material Safety Data Sheet